PROJECT SCHOLARX

# AI-ENABLED RESEARCH TOOLKIT

## Tools, workflows, checklists and integrity controls

RESEARCH TOOLKIT



AI can make you faster.
Only you can make your research meaningful.

Project ScholarX • Sprint 4 Companion Resources

akhil2453.github.io/akhil-portfolio

# Use Tools by Research Task, Not by Hype

The most reliable AI-enabled workflow separates discovery, verification, organisation, analysis, writing and governance. No single application should be treated as the research environment, evidence base and author at the same time. Features and access models change frequently; this toolkit reflects publicly documented capabilities available in July 2026 and should be rechecked before each cohort or webinar.

**AKHIL TIP #4** The right stack is usually smaller than the impressive stack. Start with one verified database, one reference manager and one carefully governed AI assistant.

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## Minimal Research Stack

A practical baseline before adding specialised tools.

| Layer | Suggested starting point | Role |
|---|---|---|
| Scholarly discovery | Google Scholar plus institutional Scopus or Web of Science access where available | Locate records; trace citations; build a defensible search. |
| Reference management | Zotero | Collect metadata and PDFs; organise; annotate; cite. |
| Source-grounded reading | NotebookLM or an institution-approved equivalent | Question and organise a bounded set of uploaded sources with citations back to those sources. |
| Literature mapping | ResearchRabbit or Connected Papers | Explore citation networks and related-paper clusters from verified seed papers. |
| Generative assistant | ChatGPT, Gemini, Claude or an institution-approved platform | Clarify, critique, structure, explain and generate code scaffolds under human verification. |
| Analysis environment | Excel, R, Python, SPSS, NVivo/ATLAS.ti or approved equivalents | Perform analysis reproducibly and retain decision records. |

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## Tool Matrix

What each tool is useful for - and what it must not replace.

| Tool | Useful for | Non-negotiable limitation |
|---|---|---|
| Google Scholar | Broad scholarly discovery; cited-by links; related works; alerts | Coverage and metadata vary; verify publisher/database record and full text. |
| Scopus | Curated abstract/citation searching, filters, | Often institutional access; database coverage |

| Tool | Useful for | Non-negotiable limitation |
|---|---|---|
| | author/source analysis | is not universal. |
| Web of Science | Curated discovery, citation tracking and research-impact workflows | Often institutional access; select databases and indexes deliberately. |
| Semantic Scholar | Free AI-powered literature search and contextual reading features | Do not treat automated summaries or relevance ranking as evidence quality. |
| Elicit | Paper search, screening, extraction and evidence-synthesis workflows | Audit the search, screening and extracted claims; read original papers. |
| ResearchRabbit | Citation-network discovery, collections and visual maps | Begin from verified seed papers; network connection is not quality appraisal. |
| Connected Papers | Similarity graph, prior works and derivative works exploration | It is a discovery graph, not a systematic search or citation tree. |
| NotebookLM | Source-grounded questions, citations, notes, reports and study artifacts from uploaded sources | Quality depends on source selection; source citations still require reading and contextual checking. |
| Zotero | Collect, organise, annotate, cite and share research sources | Imported metadata can contain errors; verify authors, titles, DOI and publication details. |
| ChatGPT | Question refinement, structured critique, coding support, web/file research and documented synthesis where available | Outputs can be incomplete or wrong; never delegate authorship, evidence verification or final judgement. |

## Official information links

**Google Scholar:** https://scholar.google.com/intl/engb/scholar/about.html

**Scopus:** https://www.elsevier.com/products/scopus

**Web of Science:** https://clarivate.com/products/scientific-and-academic-research/research-discovery-and-workflow-solutions/webofscience-platform/

**Semantic Scholar:** https://www.semanticscholar.org/product

**Elicit:** https://elicit.com/solutions/reports

**ResearchRabbit:** https://www.researchrabbit.ai/

**Connected Papers:** https://www.connectedpapers.com/about

**NotebookLM:** https://support.google.com/notebooklm/answer/16164461

**Zotero:** https://www.zotero.org/support/quick_start_guide

**ChatGPT:** https://help.openai.com/en/articles/10500283-deep-research-faq

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# Research Lifecycle Workflow

A human-governed command centre.

| Stage | Assistant role | Scholar decision gate |
|---|---|---|
| 1. Problem | AI: clarification questions, competing explanations, scope options | Human: choose a consequential problem and own the purpose. |
| 2. Search preparation | AI: keywords, synonyms, Boolean drafts, search log template | Human: test syntax, document databases and evaluate coverage. |
| 3. Discovery | Databases and mapping tools: identify records and networks | Human: select legitimate sources and avoid popularity bias. |
| 4. Reading | Source-grounded tools: extraction, comparison and questioning | Human: read original text, check context and assess quality. |
| 5. Framework and method | AI: compare alternatives and stress-test alignment | Human: justify philosophy, design, sampling, instruments and ethics. |
| 6. Data and analysis | AI: code scaffolds, cleaning checklists and interpretation questions | Human: protect data, validate code, check assumptions and own analysis. |
| 7. Writing | AI: structure critique, clarity editing and counterargument prompts | Human: produce original argument, accurate citations and calibrated claims. |
| 8. Communication | AI: audience adaptation, rehearsal questions and visual ideas | Human: defend contribution, limitations and implications. |
| 9. Audit | AI-use log, citation verification, disclosure and policy check | Human: final accountability and compliance. |

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# The Three-Layer Verification Protocol

Never move directly from AI output to thesis text.

| Layer | Question | Action |
|---|---|---|
| Existence | Does the source, dataset, instrument, policy or fact actually exist? | Locate it in a legitimate database, official site or original record. |
| Accuracy | Do metadata, quotation, statistic and interpretation match the source? | Read the relevant original passage and cross-check details. |
| Fitness | Is the source credible and appropriate for this claim, context and method? | Evaluate quality, relevance, recency, limitations and transferability. |

**CITATION RULE** A correctly formatted reference can still be fabricated, miscited or inappropriate. Formatting is the final step, not verification.

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# Responsible AI Traffic Light

A practical decision aid - always subordinate to institutional policy.

| Zone | Illustrative practices |
|---|---|
| GREEN - usually low risk | Generating keywords; explaining a concept; formatting your own notes; grammar review; creating synthetic examples; coding scaffolds with validation. |
| AMBER - supervision required | Summarising papers; coding qualitative data; drafting instruments; interpreting outputs; translating text; generating literature maps; revising academic prose. |
| RED - unacceptable or high risk | Inventing citations or data; uploading confidential identifiable material without approval; submitting generated analysis as your own; hiding material AI use; letting AI make ethics or methodology decisions. |

The colour can change with context. An apparently “green” activity may become red if it violates a university rule, data agreement, publisher policy or research-ethics approval.

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# Confidentiality and Data Safety Checklist

Stop before uploading any document, transcript or dataset.

- ☐ Is the platform approved by the university or organisation for this data class?
- ☐ Is the content confidential, commercially sensitive, privileged, copyrighted or unpublished?
- ☐ Can the task be completed with de-identified, aggregated, synthetic or placeholder data?
- ☐ Can I document exactly what was uploaded and delete it when required?
- ☐ Does the material contain names, identifiers, contact details or indirect identifiers?
- ☐ Does consent or ethics approval permit processing by this system and jurisdiction?
- ☐ Have I checked retention, training, sharing and human-review settings?
- ☐ Would I be comfortable explaining this upload to participants, my supervisor and the ethics committee?

**DEFAULT RULE** When uncertain, do not upload. Use a synthetic example and obtain formal guidance.

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# AI Use Log and Disclosure

Create an audit trail before you need one.

| Field | Record |
|---|---|
| Date and stage | When and where in the research lifecycle the tool was used. |
| Tool and access mode | Product, model/version when visible, account/workspace and relevant settings. |
| Purpose | Why the tool was used and what task it supported. |
| Input | Type of material provided; note whether synthetic, public, de-identified or confidential. |
| Output retained | What was accepted, edited, rejected or used only for orientation. |
| Verification | Sources checked, code tests, methodological review or human validation performed. |
| Disclosure requirement | University, publisher, supervisor or ethics requirement and where disclosed. |

## Adaptable Disclosure Template

**TEMPLATE** During [stages], the author used [tool and version, where available] for [specific purposes].

Outputs were reviewed, revised and verified against [original sources / validated data / software documentation]. The tool was not used to generate or fabricate data, references or findings. All research decisions, interpretations, final wording and responsibility remain with the author. This statement must be adapted to the applicable university, ethics and publisher policies.

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# Quality-Control Checklists

Run these before supervisor review or submission.

## Source and Citation Audit

☐ Every cited item exists and has been opened in a legitimate source.

☐ Author, year, title, journal/book, volume, issue, pages and DOI are correct.

☐ Every quotation matches the source and location.

☐ The cited source actually supports the sentence.

☐ Secondary citations are identified and handled appropriately.

☐ Retracted, corrected or disputed sources have been checked.

☐ Reference-manager duplicates and imported metadata errors are resolved.

☐ No AI-generated reference was retained without full verification.

## Argument and Writing Audit

☐ Claims match the strength of the evidence and design.

☐ Description is followed by comparison, evaluation or synthesis.

☐ Counterevidence and limitations are represented fairly.

☐ The contribution is not promised more strongly than the findings permit.

☐ Recommendations are traceable to evidence and context.

☐ Terminology, objectives and research questions remain consistent.

☐ The final wording is original and can be defended by the author.

☐ Material AI assistance is disclosed where required.

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# What Current Guidance Consistently Emphasises

Apply the strictest relevant policy.

| Principle | Practical meaning |
|---|---|
| Human accountability | AI systems cannot take authorship responsibility; the human author remains accountable for accuracy, integrity and |

| Principle | Practical meaning |
|---|---|
| | originality. |
| Disclosure | Many publishers and institutions require disclosure of whether and how AI-assisted technologies were used. |
| Privacy and rights | Protect personal data, confidential material, copyright and participant expectations. |
| Verification | AI outputs may be inaccurate, incomplete or biased; source and method checking cannot be delegated. |
| Local policy first | University, ethics, funder, employer and publisher rules may differ and change; check them before each use. |

Reference guidance: UNESCO Guidance for Generative AI in Education and Research

ICMJE recommendations on AI use by authors

Elsevier generative AI policies for journals

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# One-Page Implementation Plan

Build a sustainable research system this week.

| Action | My choice / date |
|---|---|
| Select the authoritative scholarly databases available through my institution. | |
| Install and organise Zotero or another approved reference manager. | |
| Create a dedicated research project/workspace with instructions and no confidential data. | |
| Write my standard verification footer and confidentiality rule. | |
| Create an AI-use log before using AI in substantive research work. | |
| Test one problem-framing prompt and one literature-search prompt. | |
| Discuss AI use, disclosure and data handling with my supervisor. | |
| Review the latest university, ethics and publisher policies. | |